

SAFI, Morocco

WMO: 601850

Lat: 32.283N

Lon: 9.233W

Elev: 45

StdP: 100.79

Time Zone: 1.00 (E01)

Period: 90-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	7.0	2.6	4.6	12.9	3.9	5.0	12.2	13.3	15.2	10.5	17.0	3.2	60	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	32.8	20.4	30.2	20.4	28.5	20.1	22.3	28.0	21.8	26.7	21.3	25.8	5.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.6	23.5	20.4	15.2	23.3	20.0	14.8	23.1	65.5	27.7	63.7	26.9	62.1	25.7	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.7	8.6	7.9	DB	3.9	38.1	1.6	4.1	2.8	41.0	1.8	43.4	1.0	45.7	-0.2	48.7
			WB	3.2	23.7	1.2	0.8	2.3	24.3	1.6	24.7	1.0	25.2	0.1	25.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	13.2	14.5	16.2	17.0	18.9	21.2	23.4	23.9	22.2	20.0	17.1	14.8
	DBStd	4.13	2.06	2.23	2.46	2.26	2.47	1.86	2.22	2.19	2.24	2.05	2.82	2.15
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	588	158	107	75	52	19	0	0	0	1	7	57	112
	CDD10.0	3123	101	127	193	211	276	337	415	431	364	309	213	147
	CDD18.3	671	0	2	10	13	36	87	157	172	116	58	20	2
	CDH23.3	3362	1	8	78	88	194	323	843	1021	502	207	93	3
(14) Wind	WSAvg	4.3	3.8	4.0	4.2	4.8	4.5	4.6	4.8	4.7	4.3	4.2	3.8	4.2
	PrecAvg	388	62	53	40	31	13	4	0	0	4	38	65	73
	PrecMax	1019	456	241	136	108	63	23	2	3	20	137	232	281
	PrecMin	137	0	0	1	0	0	0	0	0	0	0	0	0
	PrecStd	161	82	46	34	26	18	6	1	1	6	40	64	67
	DB	22.7	24.6	28.3	28.8	32.6	32.8	38.4	36.0	32.4	29.5	29.3	23.1	18.7
	MCWB	15.5	16.2	16.5	17.6	18.7	20.8	21.5	21.0	19.6	18.8	17.8	14.6	14.6
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	20.0	22.3	25.4	25.9	28.3	28.5	32.1	32.5	29.6	27.3	25.8	20.9	17.7
	2% MCWB	15.4	15.9	16.4	16.9	18.1	20.0	21.0	21.3	20.5	19.0	17.6	15.2	15.2
	5% DB	18.7	20.8	23.3	23.4	25.4	26.5	29.2	29.6	27.6	25.5	23.2	19.7	17.7
	5% MCWB	14.7	15.3	15.9	16.3	17.7	19.5	20.9	21.1	20.0	18.4	17.3	15.6	15.6
	10% DB	17.7	19.2	21.4	21.7	23.1	24.9	27.4	27.8	26.0	23.9	21.5	18.8	18.8
	10% MCWB	14.1	15.0	15.6	15.7	17.3	18.8	20.4	20.7	19.8	18.4	17.0	15.6	15.6
	WB	18.1	18.6	18.8	19.3	20.5	21.5	23.2	23.2	22.4	21.4	21.1	18.7	18.7
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	19.8	21.6	24.2	26.3	27.9	28.3	31.7	30.5	26.8	24.7	23.1	19.5	19.5
	2% WB	16.8	17.1	17.7	17.9	19.3	20.7	22.1	22.4	21.8	20.6	19.9	17.7	17.7
	2% MCDB	18.9	19.9	22.3	22.5	24.7	26.0	28.2	28.3	25.8	23.4	22.0	19.0	19.0
	5% WB	15.6	16.1	17.0	17.1	18.6	20.1	21.5	21.9	21.3	20.1	18.6	16.9	16.9
	5% MCDB	17.8	19.5	21.0	21.5	23.1	24.6	26.8	26.9	24.9	22.9	22.0	18.5	18.5
	10% WB	14.6	15.3	16.3	16.5	18.0	19.6	21.0	21.4	20.8	19.5	17.7	16.1	16.1
	10% MCDB	16.9	18.6	20.2	20.5	21.9	23.7	25.8	26.0	24.2	22.3	21.1	18.1	18.1
(35) Mean Daily Temperature Range	MDBR	9.2	9.4	9.2	8.9	7.7	7.2	7.4	7.7	7.4	8.0	8.8	8.1	8.1
	5% DB	10.4	11.5	13.1	12.2	12.0	10.3	11.4	11.4	9.9	10.9	10.9	9.1	9.1
	5% WB	6.2	6.1	6.6	5.4	4.3	3.5	3.2	3.2	3.4	4.4	5.1	5.2	5.2
	MCDBR	7.8	9.9	10.6	9.7	8.9	8.4	9.1	8.6	7.3	7.8	7.9	5.8	5.8
	MCWBR	5.4	5.9	6.3	4.9	3.9	3.4	3.1	3.0	3.0	3.8	4.5	4.2	4.2
(40) Clear-Sky Solar Irradiance	taub	0.333	0.361	0.415	0.410	0.432	0.464	0.515	0.512	0.444	0.407	0.369	0.344	0.344
	taud	2.486	2.393	2.212	2.259	2.216	2.130	1.970	2.003	2.217	2.326	2.414	2.473	2.473
	Ebn at Noon	888	891	862	882	861	829	785	781	824	828	834	852	852
	Edh at Noon	88	107	137	136	144	156	183	174	135	113	95	86	86
(44) All-Sky Solar Radiation	RadAvg	3.32	4.21	5.43	6.55	7.24	7.63	7.62	6.93	5.87	4.42	3.45	3.00	3.00
	RadStd	0.18	0.28	0.29	0.30	0.32	0.27	0.24	0.22	0.21	0.24	0.23	0.20	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.31	N/A	-0.87	+0.87	N/A	N/A	N/A	N/A	N/A	+115	+132

Nomenclature: See separate page